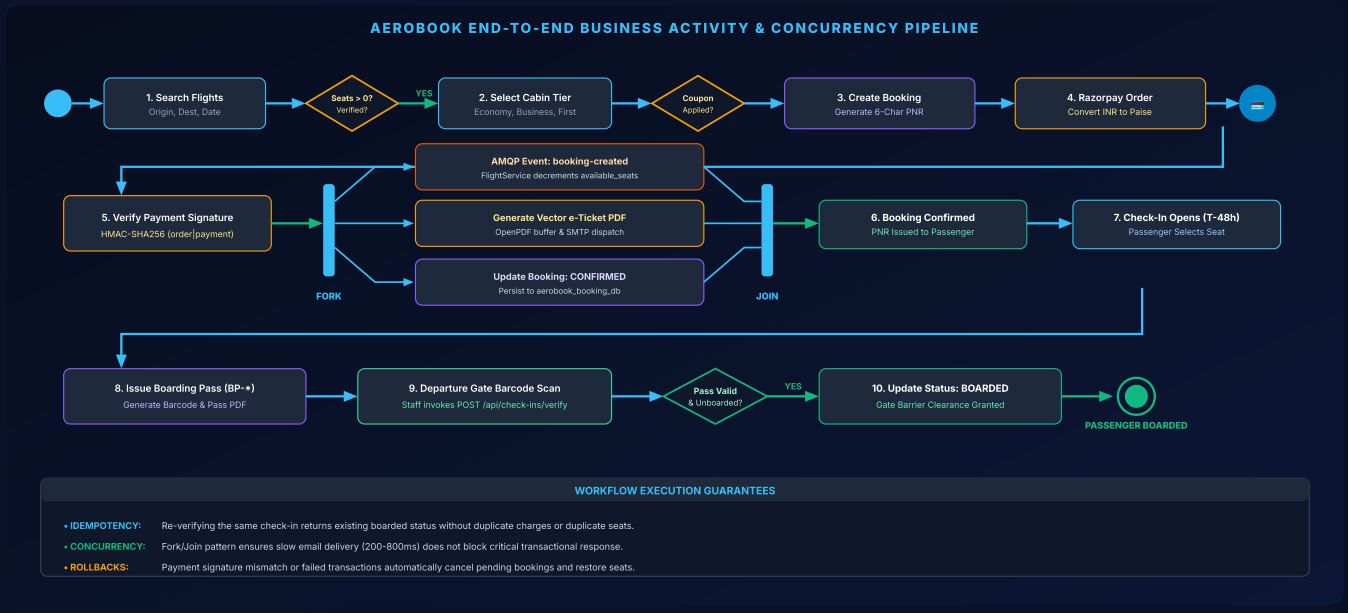
AeroBook

End-to-End Business Activity & Lifecycle Workflow

Procedural Business Execution: Flight Search, Coupon Validation, Razorpay Payment, AMQP Sync & Gate Clearance



WORKFLOW EXECUTION GUARANTEES

- **IDEMPOTENCY:** Re-verifying the same check-in returns existing boarded status without duplicate charges or duplicate seats.
- **CONCURRENCY:** Fork/Join pattern ensures slow email delivery (200-800ms) does not block critical transactional response.
- **ROLLBACKS:** Payment signature mismatch or failed transactions automatically cancel pending bookings and restore seats.

SPEAKER TALKING POINTS

- Validation Checkpoints: Seat availability, coupon discount eligibility, and HMAC-SHA256 signature checking.
- Fork/Join Concurrency: Successful payment concurrently publishes AMQP seat reduction, renders e-ticket PDF, and confirms booking.
- Gate Clearance Handshake: Airport staff mobile barcode scan transitions passenger records to BOARDED at departure gate.